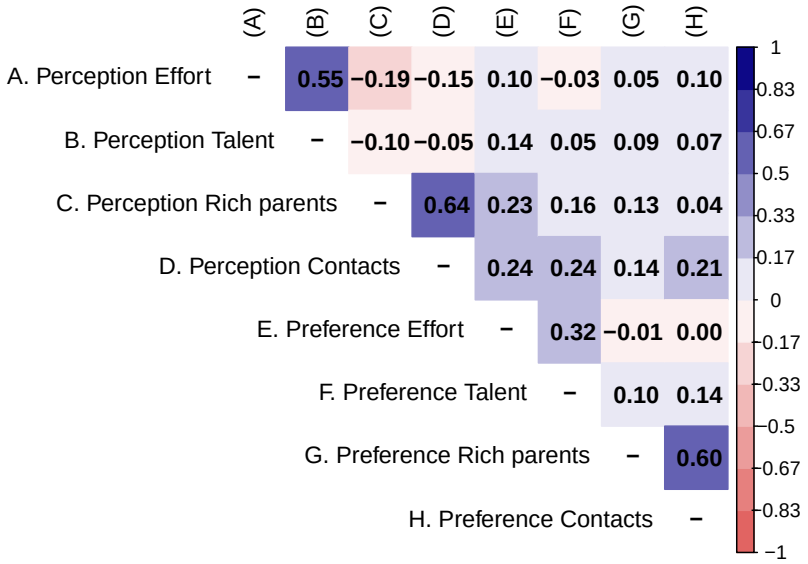
Supplementary material

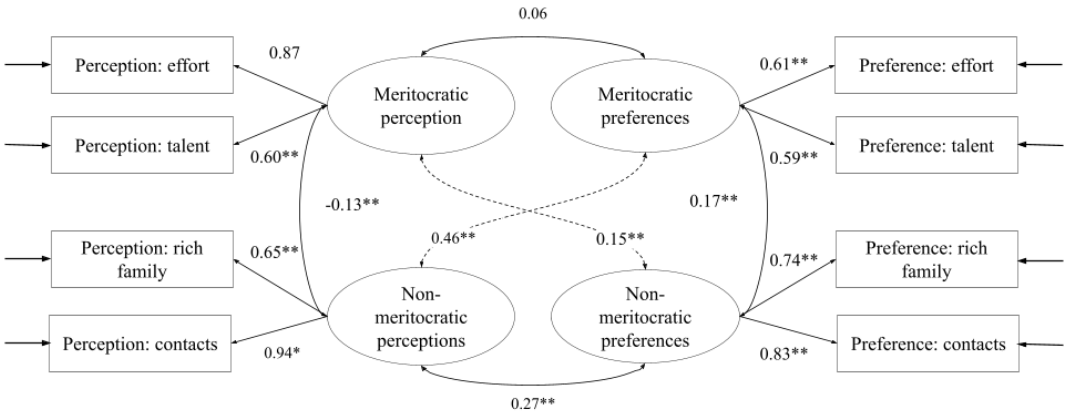
1 Correlation matrix

Figure 1: Correlation matrix for the pooled sample



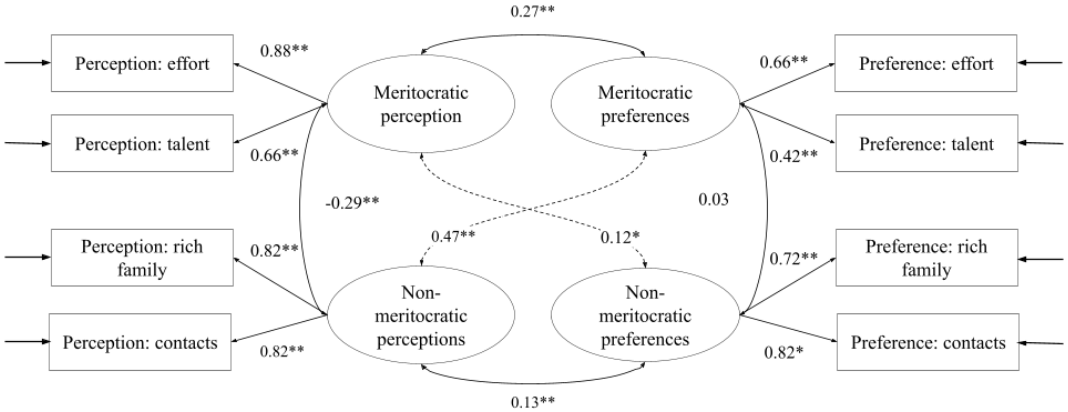
2 Confirmatory factor analysis of the meritocracy scale

Figure 2: CFA model for the first wave



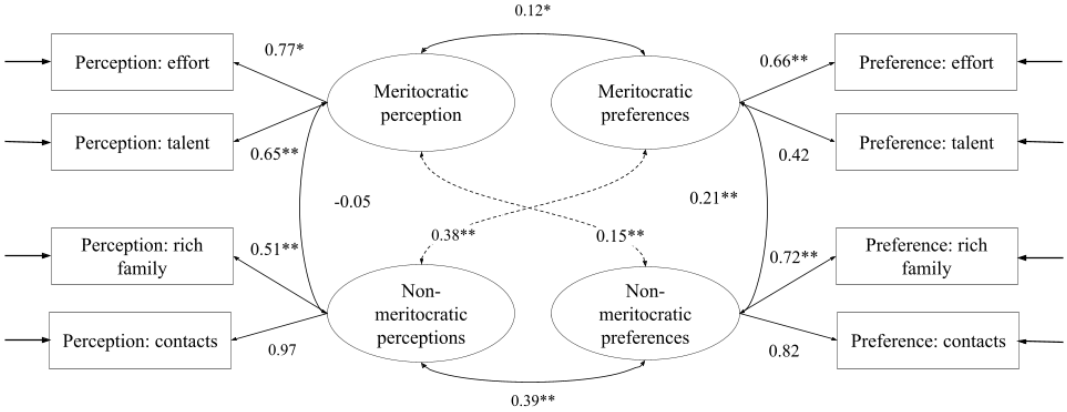
WLSMV Estimation, completely standardized solution, N=846
Chi2=62.700* / df=14, CFI=0.975, TLI=0.950, RMSEA=0.064**
p<0.01, *p<0.05

Figure 3: CFA model for the second wave



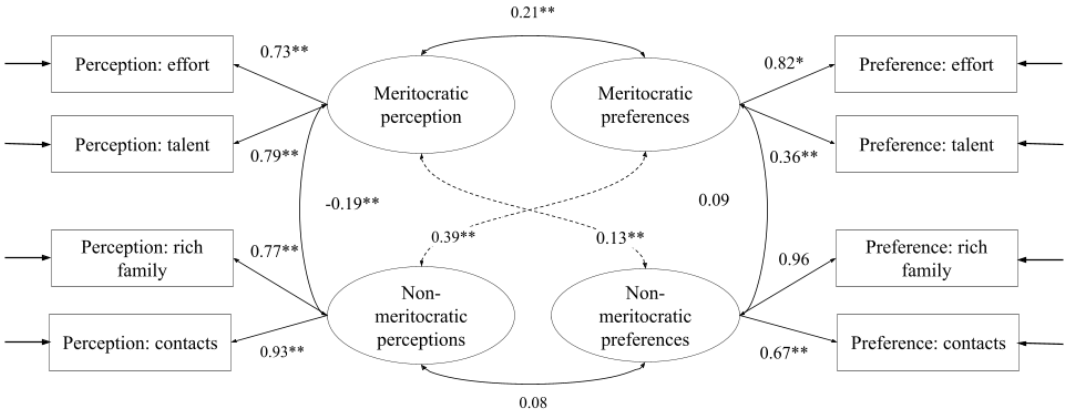
WLSMV Estimation, completely standardized solution, N=662
Chi2=69.147* / df=14, CFI=0.967, TLI=0.935, RMSEA=0.077**
p<0.01, *p<0.05

Figure 4: CFA model for the primary cohort



WLSMV Estimation, completely standardized solution, N=725
Chi2=89.168* / df=14, CFI=0.945, TLI=0.90, RMSEA=0.086**
p<0.01, *p<0.05

Figure 5: CFA model for the secondary cohort



WLSMV Estimation, completely standardized solution, N=783
Chi2=24.473* / df=14, CFI=0.995, TLI=0.991, RMSEA=0.031**
p<0.01, *p<0.05

3 Sources of misfit in longitudinal invariance

For the longitudinal strict invariance model (the more strict model), score tests provided no evidence of localized misfit. The global score test was non-significant, $\chi^2(20) = 11.93, p = .918$, and none of the univariate score tests for equality constraints was significant (all $p > .05$; smallest $p = .065$). This indicates that no single equality constraint notably contributed to the misfit. There are also no indicators of possible sources of misfit in the previous nested models (weak and strong).

Table 1: Sources of misfit in longitudinal invariance

Item	Threshold	χ^2	df	p-value	p-value adjustement BH
Perception effort	t2	10.115134	1	0.0014706	0.0352936
Perception effort	t1	7.546618	1	0.0060123	0.0721474
Perception contact	t1	5.179947	1	0.0228490	0.1464457
Preference effort	t2	5.012861	1	0.0251597	0.1464457
Perception talent	t1	4.680351	1	0.0305095	0.1464457
Perception rich parents	t1	3.516226	1	0.0607707	0.2430828
Preference effort	t3	2.177777	1	0.1400165	0.4800567
Perception rich parents	t2	1.697639	1	0.1925971	0.5332262
Perception talent	t3	1.642668	1	0.1999598	0.5332262
Perception effort	t3	1.026312	1	0.3110265	0.7233287

4 Sources of misfit between cohort invariance

In contrast, score tests for the cross-cohort invariance constraints indicated localized misfit. The largest modification indices/score contributions clustered in the perceived effort, suggesting that equality constraints on thresholds/loadings for these items are not supported across school levels. Table 1 lists the ten constraints with the largest score statistics.

Table 2: Conditional longitudinal invariance by cohort level

Model	χ^2 (df)	CFI	RMSEA (90% CI)	$\Delta\chi^2$ (Δdf)	ΔCFI	ΔRMSEA	Decision
Configural	122.27 (76)	0.990	0.032 (0.021-0.042)	0 (0)	0.000	0.000	Reference
Weak	126.95 (80)	0.990	0.032 (0.021-0.042)	4.684 (4)	0.000	-0.001	Accept
Strong	128.55 (88)	0.991	0.028 (0.017-0.038)	1.598 (8)	0.001	-0.004	Accept
Strict	131.64 (92)	0.991	0.027 (0.016-0.037)	3.086 (4)	0.000	-0.001	Accept

5 Conditional longitudinal invariance by cohort

Despite the lack of cohort measurement invariance, we examined whether plausible cohort-related heterogeneity could compromise longitudinal invariance by re-estimating the longitudinal invariance sequence while controlling for cohort (dummy-coded) as a covariate predicting each latent factor (see Table 2). Model fit remained good across all steps—from the configural baseline to the strict model—and consistently met conventional standards; for instance, the strict invariance model with cohort control showed $\chi^2(92) = 131.64$, CFI = .991, and RMSEA = .027. Importantly, including cohort as a predictor did not materially alter model fit or the invariance conclusions, suggesting that the equality constraints over time are tenable even after accounting for cohort differences in average latent factor levels.

This result has two main implications. First, it strengthens the claim that the measurement properties of the scale are temporally stable within individuals: observed over-time comparisons are not driven by shifts in how primary versus secondary students use the response categories, but reflect stability (or change) in the underlying latent constructs. Second, it clarifies the scope of the cohort non-invariance finding. Non-invariance across cohorts appears to be primarily a cross-sectional comparability problem—limiting direct comparisons of latent means or observed scores between educational levels at a given wave—rather than a threat to longitudinal analyses pooling both cohorts. Therefore, analyses of within-person change and over-time associations can be interpreted with greater confidence, while cohort contrasts should be treated cautiously (or handled via partial invariance, alignment, or cohort-specific models).